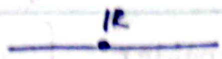
Chapter 03

Functions of Several variables

$\mathbb{R}$: The set of real numbes

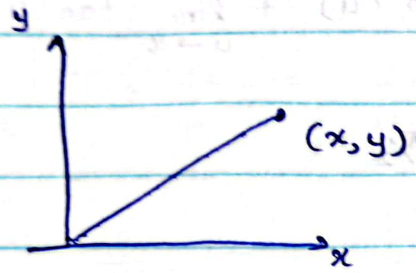


$\mathbb{R}^n = \{ (x_1, x_2, x_3, \ldots, x_n) / x_i \in \mathbb{R} \text{ for } i = 1, 2, 3 \ldots n \}$

A typical element $x = (x_1, x_2, \ldots, x_n)$ in $\mathbb{R}^n$ is called an ordered n-tuple.

when $n = 2$

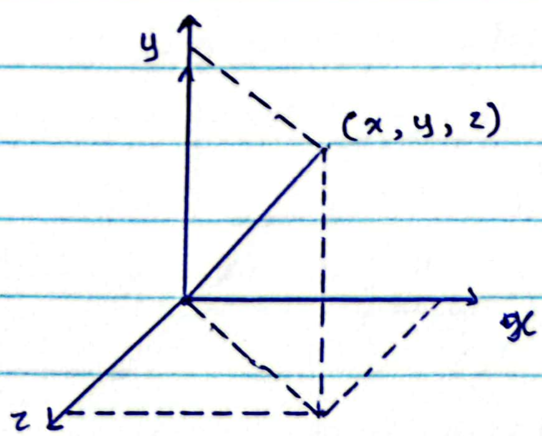
$\mathbb{R}^2 = \{(x_1, x_2) \mid x_1, x_2 \in \mathbb{R}\} = \{(x, y) \mid x, y \in \mathbb{R}\}$



← Geometrical representation of $\mathbb{R}^2$

when $n = 3$

$\mathbb{R}^3 = \{(x_1, x_2, x_3) \mid x_1, x_2, x_3 \in \mathbb{R}\}$
$= \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$



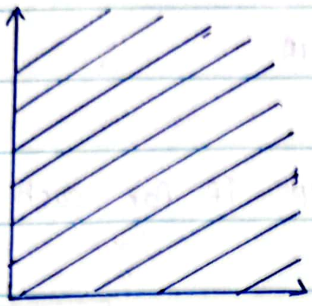
← Geomatrical Theory nation of $\mathbb{R}^3$

In this chapter, we shall stu[dy] fu[nctions]

$f : D \to \mathbb{R}$, where $D \subseteq \mathbb{R}^2$ or $D \subseteq \mathbb{R}^3$

Example 1

Let $D = \{(x,y) \mid x>0,\ y>0\}$



Define $f: D \to \mathbb{R}$ by $f(x,y) = x$

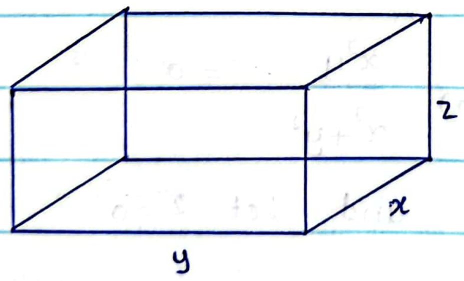
This gives the area of the unbounded region D.

Example 2

Let $D = \{(x,y,z) \mid x>0,\ y>0,\ z>0\}$

Define $v: D \to \mathbb{R}$ by $v(x,y,z) = x, y, z$

This gives the volume of unbounded cube.



The limit of a function

Note that : Let $f: \mathbb{R} \to \mathbb{R}$.

Then $\lim_{x \to a} f(x) = \ell \in \mathbb{R}$ if for each $\varepsilon > 0$ there exists $f(\varepsilon) > 0$ such that

$|f(x) - \ell| < \varepsilon$ whenever $0 < |x - a| < f(\varepsilon)$

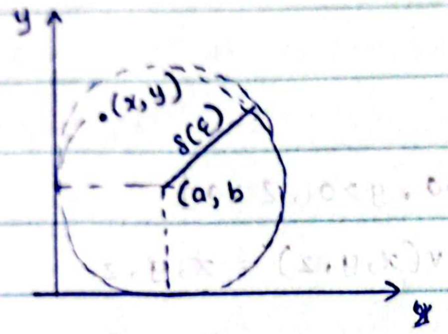
i.e., $f(x)$ can be made ~~abr~~ arbitrary close to ℓ by making sufficiently close to a but not $x = a$

Definision

let $D \subseteq \mathbb{R}^2$ and $f : D \to \mathbb{R}$

Then $\lim_{(x,y)\to(a,b)} f(x,y) = \ell \in \mathbb{R}$ if for each $\varepsilon > 0$, $\exists\ \delta(\varepsilon) > 0$ such that

$|f(x,y) - \ell| < \varepsilon$ whenever $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$



Example 3

Prove that $\lim_{(x,y)\to(0,0)} \dfrac{x^2 y}{x^2+y^2} = 0$

$|y| \le \sqrt{x^2+y^2}$

Let $f(x,y) = \dfrac{x^2 y}{x^2+y^2}$ and let $\ell = 0$

The domain of $f = D(f) = \{(x,y) \mid (x,y) \neq (0,0)\} = \mathbb{R}^2 \setminus \{(0,0)\}$

Take any $\varepsilon > 0$

consider $|f(x,y) - \ell|$

$|f(x,y) - \ell| = \left| \dfrac{x^2 y}{x^2+y^2} - 0 \right| = \left| \dfrac{x^2 y}{x^2+y^2} \right| = \dfrac{x^2}{x^2+y^2} |y|$

$\le |y|$ since $\dfrac{x^2}{x^2+y^2} \le 1$ for $(x,y) \neq (0,0)$

$\le \sqrt{x^2+y^2}$

$< \varepsilon$ whenever $\sqrt{x^2+y^2} < \varepsilon$

$y^2 \le x^2+y^2$

$|y|^2 \le x^2+y^2$

$\sqrt{|y|^2} \le \sqrt{x^2+y^2}$

i.e ; $|f(x,y) - 0| < \varepsilon$ whenever $0 < \sqrt{(x-0)^2 + (y-0)^2} < \varepsilon$

we make take $\delta(\varepsilon) = \varepsilon$

$\therefore\ |f(x,y) - 0| < \varepsilon$ whenever $0 < \sqrt{(x-0)^2 + (y-0)^2} < \delta(\varepsilon)$

Then by definition we get

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^2+y^2} = 0$$

H.ω.

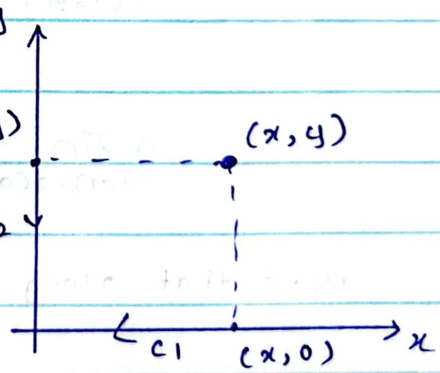
prove that $\displaystyle\lim_{(x,y)\to(0,0)} \frac{3xy^2}{x^2+y^2} = 0$

Example 4

Show that :

$\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$ does not exist

$D(f) = \mathbb{R}^2 \setminus \{0,0\}$



Along C_1 : $x > 0$ and $y = 0$

$$f(x,y) = \frac{x^2 - 0^2}{x^2+0^2} = 1$$

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} 1 = 1 \quad \text{—①}$$

Along C_2 : $y > 0$ and $x = 0$

$$f(x,y) = \frac{0^2 - y^2}{0+y^2} = -1$$

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} -1 = -1 \quad \text{——} (2)$$

$\therefore$ By (1) and (2) we conclude that

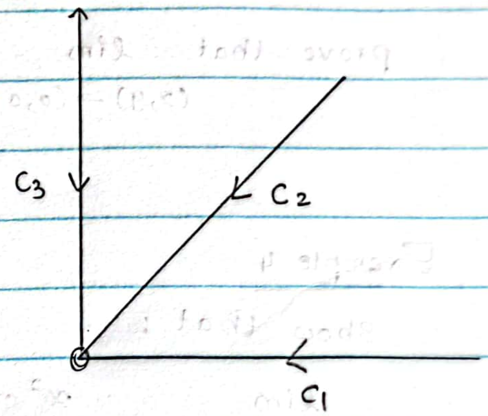
$$\lim_{(x,y)\to(0,0)} f(x,y) \text{ DNE.}$$

Example 05

Show that $\displaystyle\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$ DNE

Let $f(x,y) = \dfrac{xy}{x^2+y^2}$

$D(f) = \mathbb{R}^2 \setminus \{(0,0)\}$



Along C_1 : $y=0$ and $x>0$

$$f(x,y) = \frac{x \times 0}{x^2+0^2} = 1 \quad \text{since } x>0$$

$$\therefore \lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} 1 = 1$$

Note that along C_3 : $x=0$ and $y>0$, you will get the same li 1 and so, we don't consider C_3

Along C_2 : $y=x$ and $x>0$

$$f(x,y) = \frac{x \cdot x}{x^2+y^2} = \frac{1}{2} \quad \text{——} (2)$$

Hence, by (1) and (2), we consider that the limit

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} \text{ DNE}$$

Example 06

Show that $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy^2}{x^2+y^2}$ DNE

Let $f(x,y) = \dfrac{xy^2}{x^2+y^4}$

The $D(f) = \mathbb{R}^2 \setminus \{(0,0)\}$

Along C_1 : $x>0$ and $y=0$

$$f(x,y) = \frac{x\times 0}{x^2+0^4} = 0$$

$$\therefore \lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} 0 = 0 \quad \text{— ①}$$

Along C_2 : $x=y^2$ and $x>0$

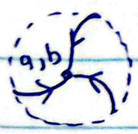
$$f(x,y) = \frac{y^2\cdot y^2}{(y^2)^2+y^4} = \tfrac{1}{2} \quad \text{since } y>0$$

$$\therefore \lim_{(x,y)\underset{C_2}{\to}(0,0)} = \lim_{x\to 0} \tfrac{1}{2} = \tfrac{1}{2} \quad \text{— ②}$$

Thus ① and ②, we consider that the limit

$$\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^4} \text{ DNE}$$

Note:-



The definition of $\lim\limits_{(x,y)\to(0,0)} f(x,y) = l$ refers only to the distance between (x,y) and (a,b). It does not refer to the direction of approve of If $\lim\limits_{(x,y)\to(a,b)} f(x,y)$ exists, no matter how (x,y)

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Definition

let $D \subseteq \mathbb{R}^2$ and $f : D \to \mathbb{R}$

f is said to be continuous at $(a,b) \in D$ if

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

i.e. i) $f(a,b)$ must be defined

ii) $\lim_{(x,y)\to(a,b)} f(x,y)$ exists

iii) $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$

Example 07

a) let $f(x,y) = \dfrac{x^2y}{x^2+y^2}$

Is f cts at $(0,0)$? No! Since f is not defined at (0

b) let $f(x,y) = \begin{cases} \dfrac{x^2y}{x^2+y} & \text{if } (x,y) \neq (0,0) \\ 3 & \text{if } (x,y) = (0,0) \end{cases}$

If f cts ab $(0,0)$?

we have already series that $\lim_{(x,y)\to(0,0)} f(x,y) = 0$

But $f(0,0) = 3$

$\therefore \lim_{(x,y)\to(0,0)} f(x,y) \neq f(0,0)$

$\therefore f$ is not cts at $(0,0)$

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Definition

let $D \subseteq \mathbb{R}^2$ and $f : D \longrightarrow \mathbb{R}$ be a function. f is said to be continuous on D if f is continuous at each point in D

Example 08

$$\text{let } h(x,y) = \begin{cases} \dfrac{x^2 y}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Since $\lim_{(x,y)\to(0,0)} h(x,y) = 0 = h(0,0)$, f is cts at $(0,0)$

In fact, f is cts on $\mathbb{R}^2$

Partial Differentiation

* Let $f(x,y) = x^2+y^2$

$\dfrac{\partial f}{\partial x} = 2x$ ⟵ Regarding y as a constant and differentiating $f(x,y)$ w.r.t. x

$\dfrac{\partial f}{\partial y} = 2y$ ⟵ Regarding x as a constant and differentiating $f(x,y)$ w.r.t. y

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* Let $f(x,y,z) = xy + yz + z^2$

$\frac{\partial f}{\partial x} = y$ ← Regarding both y and z as constant an[d] differentiating w.r.t. x

$\frac{\partial f}{\partial y} = x + z$ ← Regarding both x and z as constant an[d] differentiating w.r.t. y

* Let $f(x,y) = e^{x^2+y^2} \sin(x+2y)$

$\frac{\partial f}{\partial x} = e^{x^2+y^2}(2x)\sin(x+2y) + e^{x^2+y^2}\cos(x+2y)$

$= e^{x^2+y^2}\left[2x\sin(x+2y) + \cos(x+2y)\right]$

$\frac{\partial f}{\partial y} = e^{x^2+y^2}(2y)\sin(x+2y) + e^{x^2+y^2}\cos(x+2y)($

$= 2e^{x^2+y^2}\left[y\sin(x+2y) + \cos(x+2y)\right]$

* let $f(x,y) = x^3 + 3x^2y + xy^7$

$\frac{\partial f}{\partial x} = 3x^2 + 6xy + y^7$.

$\frac{\partial^2 f}{\partial x} = 6x + 6y = 6(x+y)$

$\frac{\partial^2 f}{\partial y . \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial y}(3x^2 + 6xy + y^7)$

$= 6x + 7y^6$

Definition

Let $D (\subseteq \mathbb{R}^n)$ be an open set and let $\underline{a} = (a_1, a_2, \ldots a_n)$ be a point in D

Let $f : D \longrightarrow \mathbb{R}$

The partial derivative of f w.r.t the i^{th} co-ordinate at $\underline{a}$ is defined to be

$$\lim_{h \to 0} \left[\frac{f(a_1, a_2, \ldots a_{i+h}, a_{i+h-1}, a_n) - f(a_1, a_2, \ldots a_i, a_{i+1} \ldots a_n)}{h} \right]$$

Provided that the limit exists. This is denoted by

$$\frac{\partial f}{\partial x_i}(\underline{a})$$

Other notation

i) $D_i(\underline{a})$

ii) $f_{x_i}(\underline{a})$

when $n = 1$

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

when $n = 2$

Let $\underline{a} = (x_0, y_0)$

$$\frac{\partial f}{\partial x}(\underline{a}) = \lim_{h \to 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

$$= D_1(\underline{a})$$

$$= f_x(x_0, y_0)$$

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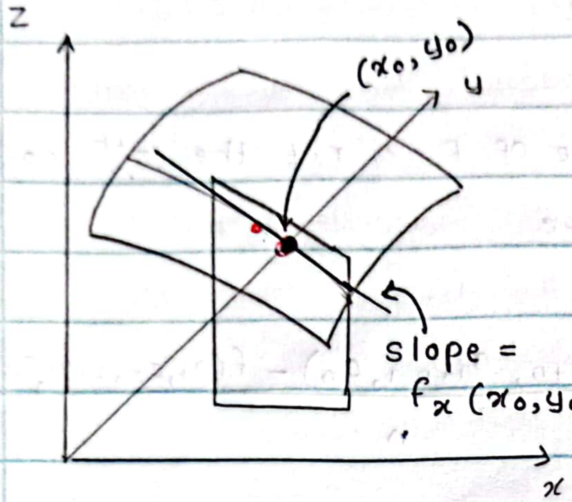
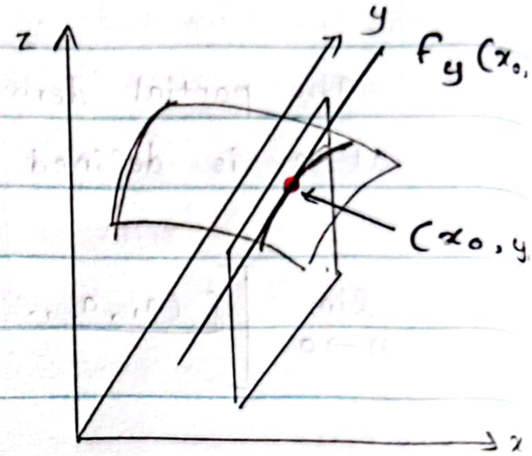
Def $\frac{\partial f}{\partial y}(\underline{a}) = \lim_{k \to 0} \frac{f(x_0, y_0 + k) - f(x_0, y_0)}{k}$

$= D_2(\underline{a}) = f_y(x_0, y_0)$





Note :

Let $z = f(x, y)$

Let Δx and Δy be increments w.r.t. x and y respect

Then $\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y)$

The differential of f (or z) denoted by df (or dz) and i

defined by

$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$ ← Total diffential of f

Example 09

let $f = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0,0) \\ 0 & \text{if } (x, y) = (0,0) \end{cases}$

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we have already seen that f is not continuous at $(0,0)$

when $(x,y) \neq (0,0)$ $f = f(x,y)$ is cts.

$$f(x,y) = \frac{xy}{x^2+y^2}$$

$$\frac{\partial f}{\partial x} = \frac{(x^2+y^2)y - xy(2x)}{(x^2+y^2)^2} = \frac{y^3 - 3x^2y}{(x^2+y^2)^2}$$

$$= \frac{y(y^2-3x^2)}{(x^2+y^2)^2}$$

Similarly

$$\frac{\partial f}{\partial y} = \frac{x(x^2-y^2)}{x^2+y^2}$$

Do $\frac{\partial f}{\partial x}(0,0)$ and $\frac{\partial f}{\partial y}(0,0)$ exist?

by definition,

$$f_x(0,0) = \frac{\partial f}{\partial x}(0,0) = \lim_{h \to 0} \frac{f(0+h,0) - f(0,0)}{h}$$

$$= \lim_{h \to 0} \frac{0-0}{h} = 0$$

$$f_y(0,0) = \frac{\partial f}{\partial y}(0,0) = \lim_{k \to 0} \frac{f(0,0+k) - f(0,0)}{k}$$

$$= \lim_{k \to 0} \frac{0-0}{k} = 0$$

Both $f_x(0,0)$ and $f_y(0,0)$ exist at $(0,0)$

but f is not continuous at $(0,0)$

i.e. the existance of the first partial derivatives at a point does not imply continuity at a point

Differentiability of functions of two variables

Let f be a function defined in some neighbourhood of (a,b). Suppose that $f_x(a,b)$ and $f_y(a,b)$ exist

Then f is said to be differentiable at (a,b) iff

$$\lim_{(h,k)\to 0,0}\left[\frac{f(a+h,b+k)-f(a,b)-hf_x(a,b)-kf_y(a,b)}{\sqrt{h^2+k^2}}\right]=0$$

Example 10

Let $f(x,y) = 2x^2y$

use the definition to show that f is differentiable at $(2,3)$

Let $a=2$ and $b=3$

$$\text{Let } R(h,k) = \frac{f(a+h, b+k)-f(a,b)-hf_x(a,b)-kf_y(a,b)}{\sqrt{h^2+k^2}} \quad \text{—(1)}$$

we must show that $\lim_{(h,k)\to(0,0)} R(h,k) = 0$

$f(x,y) = 2x^2y \Rightarrow f_x(x,y) = 4xy, \quad f_y(x,y) = 2x^2$

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Hence, $f_x(2,3) = 24$ and $f_y(2,3) = 8$

Also, $f(2,3) = 24$ and $f(2+h, 3+k) = 2(2+h)^2(3+k)$

Now by ① we get

$$R(h,k) = \frac{2(2+h)^2(3+k) - 24 - h\times 24 - 8k}{\sqrt{h^2+k^2}}$$

$$= \frac{8hk + 2h^2k + 6h^2}{\sqrt{h^2+k^2}}$$

$$= \frac{8hk}{\sqrt{h^2+k^2}} + 2\frac{h^2k}{\sqrt{h^2+k^2}} + \frac{6h^2}{\sqrt{h^2+k^2}}$$

$$\lim_{(h,k)\to(0,0)} R(h,k) = 8\lim_{(h,k)\to(0,0)} \frac{hk}{\sqrt{h^2+k^2}} + 2\lim_{(h,k)\to(0,0)} \frac{h^2k}{\sqrt{h^2+k^2}}$$

$$+ 6\lim_{(h,k)\to(0,0)} \frac{h^2}{\sqrt{h^2+k^2}} \quad \text{—②}$$

Take any $\varepsilon > 0$

Now, $\left|\frac{hk}{\sqrt{h^2+k^2}} - 0\right| = \left|\frac{hk}{\sqrt{h^2+k^2}}\right| = |h| \frac{|k|}{\sqrt{h^2+k^2}} \le |h| \le \sqrt{h^2+k^2}$

$\therefore \lim_{(h,k)\to(0,0)} \frac{hk}{\sqrt{h^2+k^2}} = 0 < \varepsilon$ whenever $0 < \sqrt{h^2+k^2} < \varepsilon$

$\delta(\varepsilon)$

Similarly, we can show that

$\lim_{(h,k)\to(0,0)} \frac{h^2k}{\sqrt{h^2+k^2}} = 0$ and $\lim_{(h,k)\to(0,0)} \frac{h^2}{\sqrt{h^2+k^2}} = 0$

Now by ②, we get $\lim_{h,k\to 0,0} R(h,k) = 0$

$\therefore f$ is differentiable at $(2,3)$

Note that f is differentiable at (a,b)

$\iff \lim_{(h,k)\to(0,0)} R(h,k) = 0$

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Theorem

If a function $f = f(x,y)$ is differentiable at (a,b) then f is continuous at (a,b)

Second order derivatives

Let $f = f(x,y)$

Suppose that $f_x(x,y)$ and $f_y(x,y)$ exist.

$$f_x = \frac{\partial f}{\partial x} \Rightarrow \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}(f_x) = f_{x\,x}$$

$$f_y = \frac{\partial f}{\partial y} \Rightarrow \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}(f_y) = f_{y\,y}$$

$$\frac{\partial^2 f}{\partial x \cdot \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}(f_y) = f_{xy}$$

$$\frac{\partial^2 f}{\partial y \cdot \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial y}(f_x) = f_{y\,x}$$

Note that, in general $f_{xy} \neq f_{yx}$

H.W

Find an example for a function $f = f(x,y)$ such that

$$\frac{\partial^2 f}{\partial x \partial y} \neq \frac{\partial^2 f}{\partial y \partial x}$$

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Example 11

Let $f(x,y) = x^3 y + e^{xy^2}$

$f(x) = \frac{\partial f}{\partial x} = 3x^2 y + e^{xy^2} \cdot y^2$

$\frac{\partial^2 f}{\partial x \cdot \partial x} = \frac{\partial}{\partial x}(f_x) = 3x^2 y + e^{xy^2} \cdot 2y\,(xy^2+1)$ — ①

$f_y = \frac{\partial f}{\partial y} = x^3 + 2yx\,e^{xy^2}$

$\frac{\partial^2 f}{\partial x \cdot \partial y} = \frac{\partial}{\partial x}(f_y)$

$= 3x^2 + 2y\left[e^{xy^2} + xy^2 e^{xy^2}\right]$

$= 3x^2 + 2y\,e^{xy^2}\left[1 + xy^2\right]$ — ②

Then it is clear that $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$

Theorem

If f_x and f_y are both differentiable at a point (a,b) of the domain of a function f, then $f_{xy}(a,b) = f_{yx}(a,b)$

Theorem

If f_y exists in a neighbourhood of a point (a,b) of the domain of the definition of a function f and f_{yx} is continuous at (a,b) then $f_{xy}(a,b)$ exists and

$f_{xy}(a,b) = f_{yx}(a,b)$

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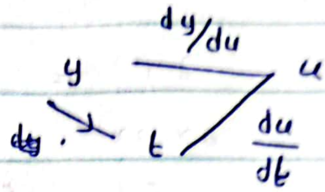
Chain Rule

If $y = f(x)$ is a differentiable function of u and $u = g$...
then,

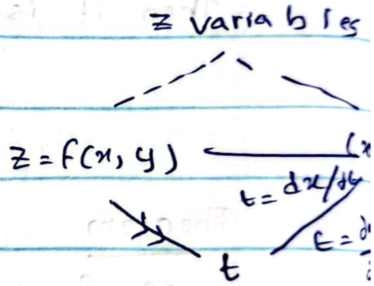
$$\frac{dy}{dt} = \frac{dy}{du} \cdot \frac{du}{dt}$$



let $z = f(x, y)$ be a differentiable function of x and y
Suppose that $x = g(t)$ and $y = h(t)$ are both differentiable function of t

Then z is a differentiable function of t and

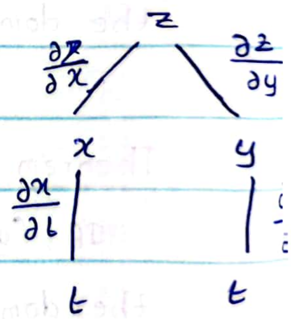
$$\frac{dz}{dt} = \frac{dz}{dx} \cdot \frac{dx}{dt} + \frac{dz}{dy} \cdot \frac{dy}{dt}$$



Example 12

Let $Z = e^{x} \sin y$, $x = t^2$ and $y = \sin t$
find dz/dt by chain rule.

$$\frac{dz}{dt} = \frac{dz}{dx} \cdot \frac{dx}{dt} + \frac{dz}{dy} \cdot \frac{dy}{dt} \quad \text{—(1)}$$



$Z = e^{x} \sin y$

$\Rightarrow \frac{\partial z}{\partial x} = e^{x} \sin y \qquad \frac{\partial z}{\partial y} = e^{x} \cos y$

$x = t^2 \longrightarrow dx/dt = 2t$

$y = \sin t \longrightarrow dy/dt = \cos t$

Now by ① we get

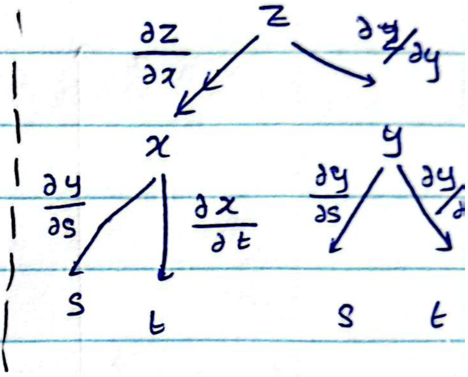
$$\frac{dz}{dt} = e^{x}\sin y \cdot 2t + e^{x}\cos y \cdot \cos t$$

$$= e^{x}(2t\sin y + \cos t . \cos y)$$

Let $z = f(x, y)$ be a differentiable function of x and y suppose that $x = g(s, t)$ and $y = h(s, t)$ when g_s, g_t, h_s and h_t all exist.

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$



Example 13

Let $z = e^{x}\sin y$ $\quad x = s \cdot t^{2}$, $y = s^{3}t$

Find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$$= (e^{x}\sin y)\,t^{2} + (e^{x}\cos y)(3s^{2}t)$$

$$= e^{x}t\,(t\sin y + 3s^{2}\cos y)$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$

$$= (e^{x}\sin y)(2ts) + (e^{x}\cos y) \cdot s^{3}$$

$$= se^{x}(2t\,\cancel{e^{x}}\sin y + s^{2}\cos y)$$

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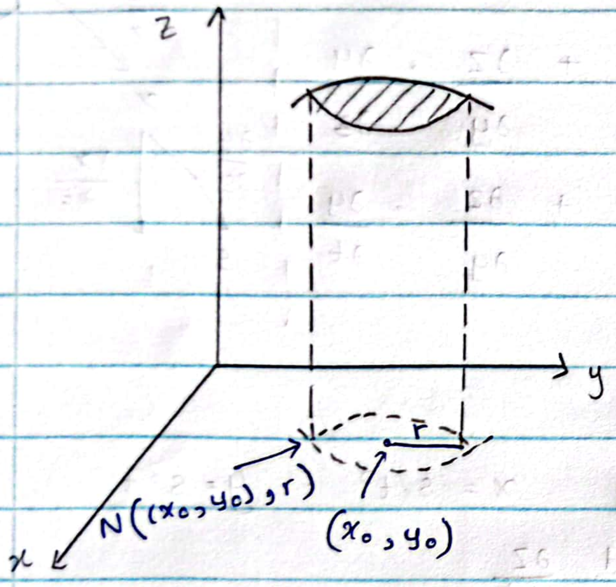
Extreme values of functions of two variables.

Let f be a real valued function of two variables defi… in a neighbourhood of a point (x_0, y_0)

This function is said to have a local maximum at (x_0, y_0) if $f(x, y) \le f(x_0, y_0)$ for all points (x, y) sufficiently closed to (x_0, y_0)

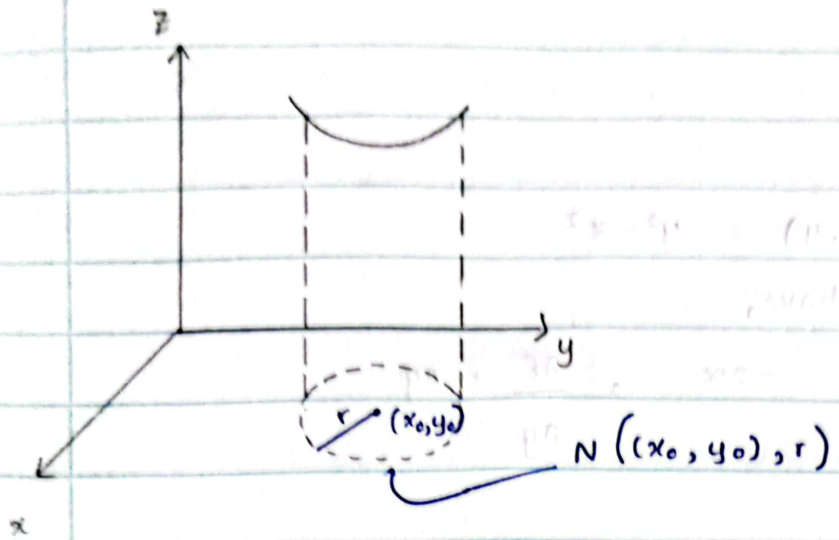


$$N((x_0, y_0), r) = \{(x, y) \in \mathbb{R}^2 \mid 0 < \sqrt{(x-x_0)^2 + y - y_0} < r\}$$

i.e. f has a local maximum at (x_0, y_0) if $\exists\, r > 0$ such…

$$f(x, y) \le f(x_0, y_0) \text{ for all } (x, y) \in N((x_0, y_0), r)$$

A local minimum is defined in a similar way



i.e. f has a local minimum at (x_0, y_0) if $\exists\, r > 0$ such that.

$$f(x_0, y_0) \leq f(x, y) \text{ for all } (x, y) \in N((x_0, y_0); r)$$

Definition

A point (x_0, y_0) where $\partial f/\partial x\,(x_0, y_0) = 0 = \dfrac{\partial f}{\partial y}(x_0, y_0)$ or at least one of $\partial f/\partial x\,(x_0, y_0)$ and $\partial f/\partial y\,(x_0, y_0)$ is undefined is called a critical point of f.

Definition

A function $z = f(x, y)$ has a saddle point at (x_0, y_0) if (x_0, y_0) is a critical point of f and for each $\varepsilon > 0$ there exist points (x_1, y_1) and (x_2, y_2) is $N((x_0, y_0), \varepsilon)$ with

$$f(x_1, y_1) < f(x_0, y_0) \text{ and } f(x_2, y_2) > f(x_0, y_0)$$

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Example 14

Let $f(x,y) = y^2 - x^2$

Then we have,

$$\frac{\partial f}{\partial x} = -2x \quad , \quad \frac{\partial f}{\partial y} = 2y$$

The critical points of f are given by

$$\frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

i.e $\quad -2x = 0 \quad$ and $\quad 2y = 0$

$\quad x = 0 \quad$ and $\quad y = 0$

So $(0,0)$ is a critical point of f

Is $(0,0)$ a saddle point of f ?

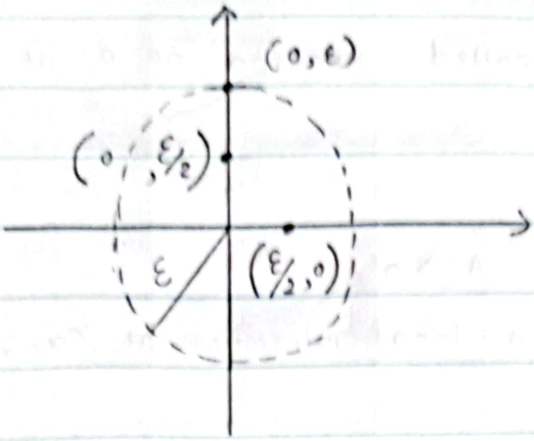
Take any $\varepsilon > 0$

If there exist $(x_1, y_1), (x_2, y_2) \in N((0,0), \varepsilon)$

Such that,

$f(x_1, y_1) < f(0,0)$ and $f(x_2, y_2) > f(0,0)$, then $(0,0)$ is a saddle point of f

$$N((0,0), \varepsilon) = \left\{ (x,y) \in \mathbb{R}^2 \mid 0 < \underbrace{\sqrt{(x-0)^2 + (y-0)^2}}_{\sqrt{x^2+y^2}} < \varepsilon \right\}$$



$f(0,0) = 0$

Note that $(\varepsilon/2, 0)$, $(0, \varepsilon/2) \in N((0,0), \varepsilon)$

Take $(x_1, y_1) = (\varepsilon/2, 0)$ and $(x_2, y_2) = (0, \varepsilon/2)$

$$f(x_1, y_1) = 0^2 - (\varepsilon/2)^2 = -\varepsilon^2/4 < 0 = f(0,0)$$

$$f(x_2, y_2) = (\varepsilon/2)^2 - 0^2 = \varepsilon^2/4 > 0 = f(0,0)$$

Therefore, f has a saddle point at $(0,0)$

Suppose that the 2nd order partial derivatives of f are continuous in a disk centered at (x_0, y_0) and that,

$$\frac{\partial f}{\partial x}(x_0, y_0) = 0 = \frac{\partial f}{\partial y}(x_0, y_0)$$

Let $A = \dfrac{\partial^2 f}{\partial x^2}(x_0, y_0)$

$B = \dfrac{\partial^2 f}{\partial y \partial x}(x_0, y_0)$

$C = \dfrac{\partial^2 f}{\partial y^2}(x_0, y_0)$ and $\Delta = AC - B^2$

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$\begin{bmatrix} A & B \\ B & C \end{bmatrix}$ is called Heasion of f at (x_0, y_0)

a) If $\Delta > 0$ and $A > 0$,

then f has a local minimum at (x_0, y_0)

b) If $\Delta > 0$ and $A < 0$,

then f has a local maximum at (x_0, y_0)

c) If $\Delta < 0$,

then f has a saddle point at (x_0, y_0)

Note :

If $\Delta = 0$, no conclusion can be drawn

i.e, the test provides no information.

Example 15

Let $f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$

The critical points of f are given by

$$\frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

$$3x^2 + 3y^2 - 6x = 0 \quad \text{and} \quad 6xy - 6y = 0$$

$$x^2 + y^2 - 2x = 0 \quad \text{and} \quad xy - y = 0$$

$$x^2 + y^2 - 2x = 0 \quad \text{and} \quad (x-1)y = 0$$

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case I

$x^2 + y^2 - 2x = 0$ and $y = 0$

$\Rightarrow x^2 - 2x = 0 \longrightarrow x(x-2) = 0 \longrightarrow x = 0$ or $x = 2$

Then the critical points of f are $(0,0)$, $(2,0)$

Case II

$x^2 + y^2 - 2x = 0$ and $x = 1$

$1 + y^2 - 2 = 0$

$y^2 - 1 = 0$

$(y-1)(y+1) = 0$

$y = 1$ or $y = -1$

Then the critical points of f are $(1,-1)$, $(1,1)$

All the critical points of f are

$(0,0)$, $(2,0)$, $(1,1)$ and $(1,-1)$

At a point (x,y)

$$A = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}(3x^2 - 3y^2 - 6x) = 6x - 6 = 6(x-$$

$$B = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}(6xy - 6y) = 6y$$

$$C = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial y}\right) = \frac{\partial}{\partial y}(6xy - 6y) = 6(x-1)$$

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critical Points	A	B	C	$\Delta = AC - B^2$	Nature of the critical point
$(0,0)$	-6	0	-6	$36>0$	Local maximum
$(2,0)$	6	0	6	$36>0$	Local minimum
$(1,1)$	0	6	0	$-36<0$	saddle point
$(1,-1)$	0	-6	0	$-36<0$	saddle point

H.W.

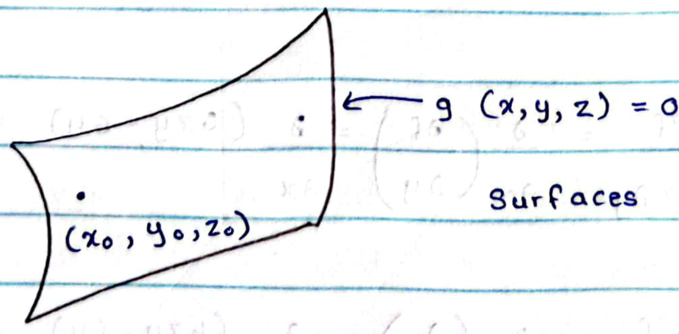
Let $f(x,y) = 8x^4 + 12x^2y - 9x^3$

Find and classify the critical point of f.

Lagrange's Multipliers

This is a method finding maxima / minima of a funct. of several variables subject to a constrain (a side con.

Suppose that we have to find the extrema (Minima or maxima) of $f = f(x,y,z)$ subject to the condition (con. $g(x,y,z) = 0$.



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Suppose that f has an extrema at (x_0, y_0, z_0) on S

Let $F = f + \lambda g$

Then we may show that

$$\Delta F(x_0, y_0, z_0) = 0$$

$\Rightarrow \nabla (f + g\lambda)(x_0; y_0, z_0) = \underline{0}$

$\Rightarrow \dfrac{\partial(f + g\lambda)}{\partial x} \underline{i} + \dfrac{\partial(f + g\lambda)}{\partial y} \underline{j} + \dfrac{\partial(f + \partial\lambda)}{\partial z} \underline{k} = \underline{0}$

$\Rightarrow (f_x + \lambda g_x) \underline{i} + (f_y + \lambda g_y) \underline{j} + (f_z + \lambda g_z) \underline{k} = \underline{0}$

Since $\underline{i}, \underline{j}$ and $\underline{k}$ are linearly independent, we have

$$f_x + \lambda g_x = 0$$
$$f_y + \lambda g_y = 0$$
$$f_z + \lambda g_z = 0$$

at (x_0, y_0, z_0) and $g(x_0, y_0, z_0) = 0$

Example 16

Find the minimum value of $f(x, y, z) = x^2 + y^2 + z^2$ subject to $2x + 3y - z = 1$

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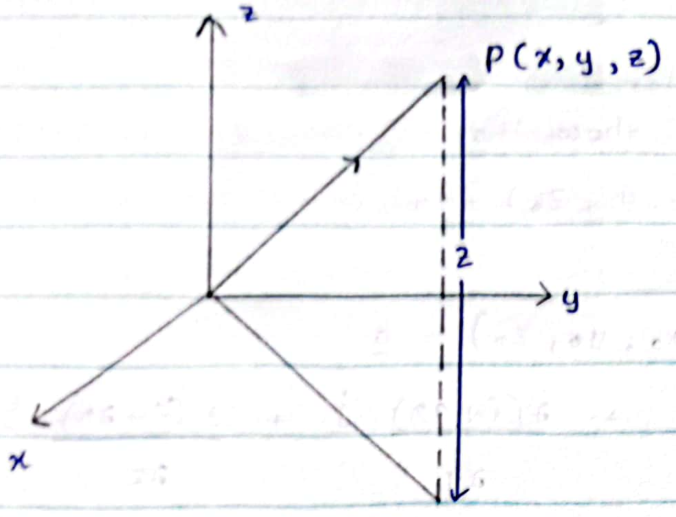
solution



$2x + 3y - z = 1$

on zoy - plane $(x=0)$;

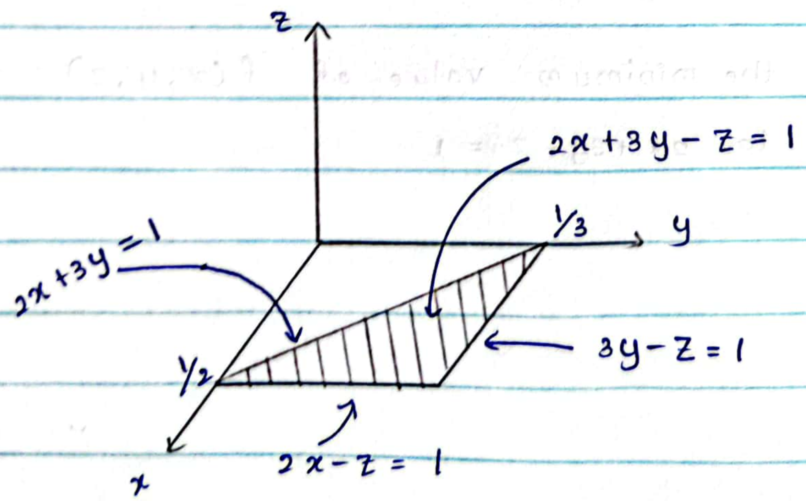
$x = 0 \Rightarrow 3y - z = 1$

on xoz - plane $(y=0)$;

$y = 0 \Rightarrow 2x - z = 2$

on xoy - plane $(z=0)$;

$z = 0 \Rightarrow 2x + 3y = 1$



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Let $g(x, y, z) = 2x + 3y - z - 1$

$f(x, y, z) = x^2 + y^2 + z^2 = (\text{distance})^2$ from the origin is to be minimized assuming that (x, y, z) lies on $g(x, y, z) = 0$

Maximum does not exist but the minimum exists.

Let $F = f + \lambda g$ for some scalar λ ($t_1(\neq 0)$)

$\Rightarrow \quad F = x^2 + y^2 + z^2 + \lambda(2x + 3y - z - 1)$

$\Rightarrow \quad F_x = 2x + 2\lambda$

$F_y = 2y + 3\lambda$

$F_z = 2z - \lambda$

Now by lagrange's multipliers, at the minimum of F (subject to $g(x, y, z) = 0$), we have:

$F_x = 2x + 2\lambda = 0 \longrightarrow x = -\lambda$ — (1)

$F_y = 2y + 3\lambda = 0 \longrightarrow y = -3\lambda/2$ — (2)

$F_z = 2z - \lambda = 0 \longrightarrow z = \lambda/2$ — (3)

we also have

$g(x, y, z) = 2x + 3y - z - 1 = 0$ — (4)

No :

Substituting x, y and z from (1), (2) and (3) respectively into (4) we have :

$$2(-\lambda) + 3(-3\lambda/2) - \lambda/2 - 1 = 0$$

$$-2\lambda - 9\lambda/2 - \lambda/2 - 1 = 0$$

$$\therefore \lambda = -1/7$$

So $x = 1/7$, $y = -3/2\,(-1/7) = 3/14$, and $z = -1/14$

Therefore, the minimum of f must occur at $(1/7, 3/14, -1/14)$

The minimum value of f

$$= f(1/7, 3/14, -1/14)$$

$$= (1/7)^2 + (3/14)^2 + (-1/14)^2$$

$$= 0.129$$

Example 17

Suppose that the temperature at a point (x, y, z) on the unit sphere $S : x^2 + y^2 + z^2 = 1$ is given by $T(x, y, z) = 30 + 5(x + z)$

Find the extreme values of T

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Solution

Let $F = 30 + 5(x+z) + \lambda(x^2+y^2+z^2-1)$, where $\lambda \neq 0$

we then have

$F_x = 5 + 2\lambda x$

$F_y = 2\lambda y$

$F_z = 5 + 2\lambda z$

Now by lagrange's multipliers, at the extreme values of we have ;

$5 + 2\lambda x = 0$ — ①

$2\lambda y = 0$ — ②

$5 + 2\lambda z = 0$ — ③

we also have :

$x^2 + y^2 + z^2 = 1$ — ④

① $\longrightarrow$ $x = -5/2\lambda$

② $\longrightarrow$ $y = 0$

③ $\longrightarrow$ $z = -5/2\lambda$, since $\lambda \neq 0$

Then ④ gives

$$\left(\frac{-5}{2\lambda}\right)^2 + 0^2 + \left(\frac{-5}{2\lambda}\right)^2 = 1$$

$$\lambda = \pm \frac{5}{\sqrt{2}}$$

Then $x = \pm \frac{1}{\sqrt{2}}$, $y = 0$ and $z = \pm \frac{1}{\sqrt{2}}$

i.e. the extreme points are $\left(-\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)$ and
$\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$

At $\left(-\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)$, $T = 30 + 5\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$

$= 30 - 5/\sqrt{2}$

and

at $\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$, $T = 30 + 5\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)$

$= 30 + 5\sqrt{2}$

<u>H.w.</u>

Find the maximum and minimum values of x^2+y^2+z

Subject to the conditions

$$\frac{x^2}{4} + \frac{y^2}{5} + \frac{z^2}{25} = 1 \quad \text{and} \quad z = x+y$$

Let $g(x,y,z) = x^2/4 + y^2/5 + z^2/25 - 1$ and

$h(x,y,z) = x + y - z$

Let $F = f + \lambda g + \mu h$

$= x^2+y^2+z^2 + \lambda\left(x^2/4 + y^2/5 + z^2/25 - 1\right) +$

$\mu(x+y-z)$;

when $\lambda, \mu \neq 0$

double / triple only.

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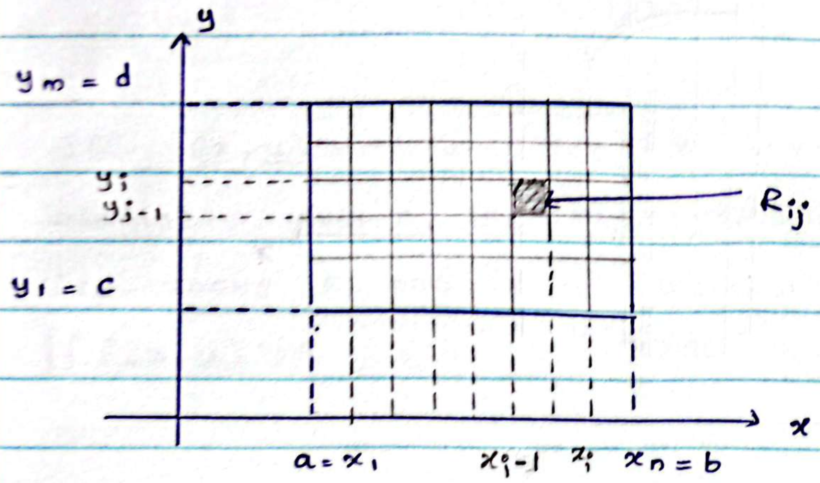
Multiple Integrals.

Let $R = \{(x,y) \in \mathbb{R}^2 \mid a \le x \le b \ , \ c \le y \le d\}$

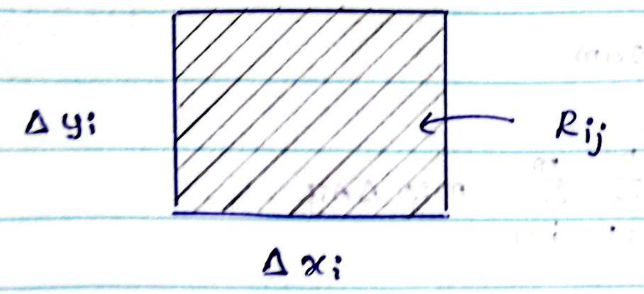
Let $f : R \longrightarrow \mathbb{R}$ be a bounded function

Consider the partition P of R into $m \times n$ rectangles created by the partitions $\{x_1, x_2, \dots x_n\}$ of $[a,b]$ and $\{y_1, y_2, \dots, y_n\}$ of $[c,d]$



Let $R_{ij} = \{(x,y) \in \mathbb{R}^2 / \ x_{i-1} \le x \le x_i \quad y_{j-1} \le y \le y_j\}$

for $i/j = 1, 2, 3, \dots$



Let $\Delta x_i = x_i - x_{i-1}$

for $i = 1, 2, \dots n$

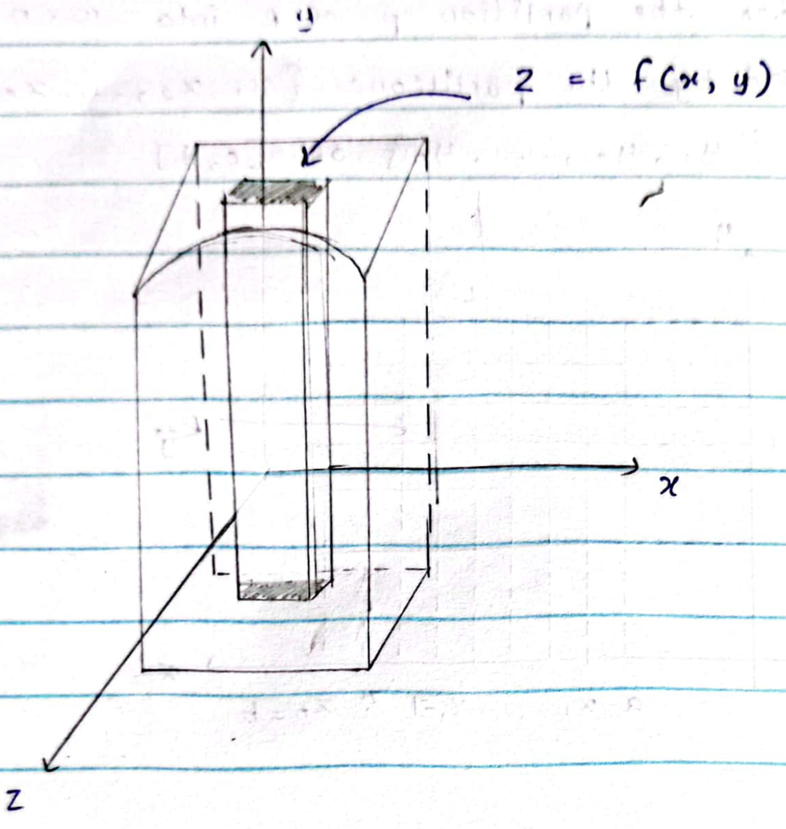
$\Delta y_i = y_{j+1} - y_i$

for $j = 1, 2, \dots n$

The ~~re~~ area of $R_{ij} = \Delta A_{ij} = \Delta x_i \times \Delta y_j$

Let $m_{ij} = \inf \{ f(x,y) \mid (x,y) \in R_{ij} \}$

$$M_{ij} = \sup \{ f(x,y) \mid (x,y) \in R_{ij} \}$$



Now form the upper sum

$$U(f,P) = \sum_{i=1}^{n} \sum_{j=1}^{m} M_{ij} \, \Delta A_{ij}$$

and the lower sum

$$L(f,P) = \sum_{i=1}^{n} \sum_{j=1}^{m} m_{ij} \, \Delta A_{ij}$$

Let $U(f) = \inf \{ U(f,P) \mid P \text{ is a particial of } R \}$ and
$L(f) = \sup \{ L(f,P) \mid P \text{ is a particial of } P \}$

If $U(f) = L(f)$, then f is said to be Riemann integrable over R.

The common value is by $\iint f(x,y)\, dx$ or $\iint f(x,y)\, dy$.

Note

If $f(x,y) \geq 0$ for $(x,y) \in \mathbb{R}$ and f is Riemann integrable over R, then the volume of the solid that lies above R and under the surface is defined by $\iint f(x,y)\, dA$

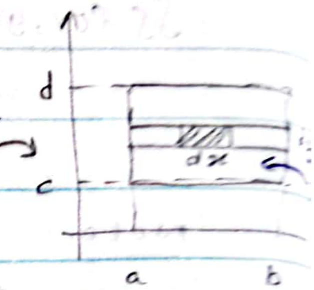
Theorem 1

Let f be Riemann integrable over the rectangle

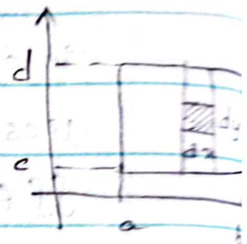
$$R = \{(x,y) \in \mathbb{R}^2 \mid a \leq x \leq b, \quad c \leq y \leq d\}$$
$$= [a,b] \times [c,d]$$

Then

$$\iint_R f(x,y)\,dA = \int_c^d \left[\int_{x=a}^{b} f(x,y)\,dx\right] dy$$



$$= \int_{x=a}^{b} \left[\int_{x=c}^{d} f(x,y)\,dy\right] dx$$



Example 1

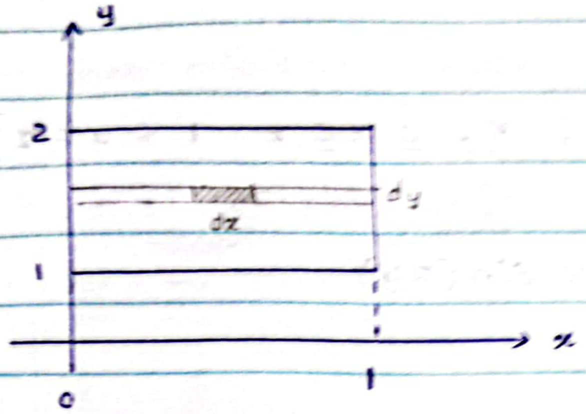
Evaluate $\iint_R (xy - x^3)\,dA$,

where

$$R = \{(x,y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, \; 1 \leq y \leq 2\}$$

Solution

Let $f(x,y) = xy - x^3$ (Integrand)



By theorem 1, we have,

$$\iint_R f(x,y)\,dA = \int_{y=1}^{2}\left[\int f(x,y)\,dx\right]dy$$

$$= \int_{y=1}^{2}\left[\int_{x=0}^{1}(xy - x^3)\,dx\right]dy$$

Integrate w.r.t. x holding y constant

$$= \int_{y=1}^{2}\left[x^2 y/2 - x^4/4\right]_{x=0}^{1} dy$$

$$= \int_{y=1}^{2}\left[y/2 - 1/4\right]dy$$

$$= \left[y^2/4 - 1/4\, y\right]_{y=1}^{2}$$

$$= 4/4 - 1/4 - 2/4 + 1/4$$

$$= 1 - 1/2$$

$$= \underline{\underline{1/2}}$$

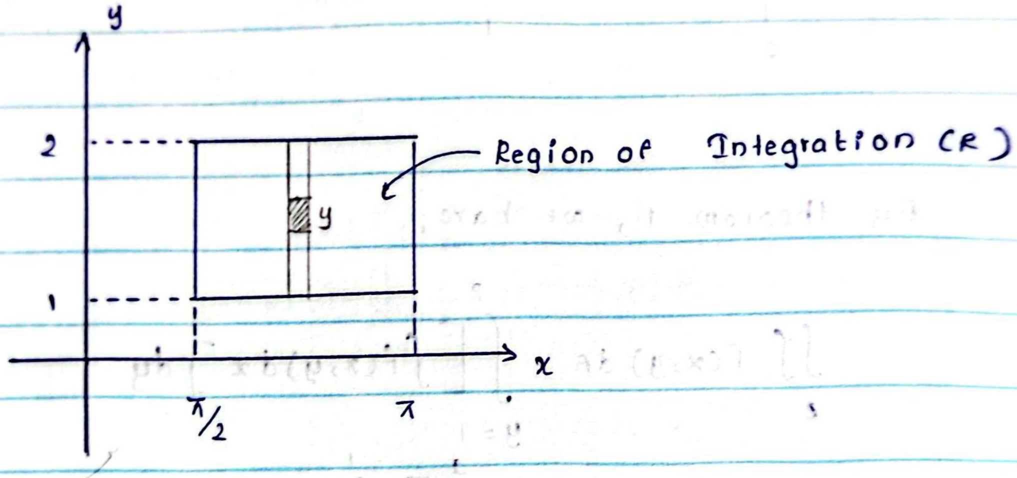
Example 2

Evaluate $\iint\limits_R x \sin(xy)\, dx\, dy$

where

$R = \{ (x, y) \in \mathbb{R}^2 \mid \pi/2 \leq x \leq \pi \quad 1 \leq y \leq 2 \}$

Let $f(x, y) = x \sin(xy)$



$$\iint\limits_R x \sin(xy)\, dx\, dy = \int_{x=\pi/2}^{\pi} \left[\int_{y=1}^{2} x \sin(xy)\, dy \right] dx$$

$$= \int_{x=\pi/2}^{\pi} -(\cos 2x - \cos x)\, dx$$

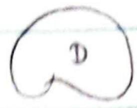
$$= - \left[\tfrac{1}{2} \sin 2x - \sin x \right] \Big|_{x=\pi/2}^{\pi}$$

$$= [(0-0)(0-1)]$$

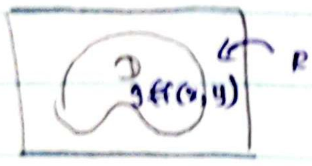
$$= \underline{\underline{-1}}$$

Double Integral over general Region.

Let D be a bounded region



Then there exists a rectangle R containing D

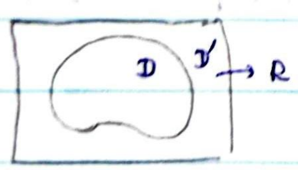


Define a new function g on R by

$$g(x,y) = \begin{cases} f(x,y) & \text{if } (x,y) \in D \\ 0 & \text{if } (x,y) \in D \end{cases}$$

If g is integrable over R, then we say that f is integrable over D. and we define the double integral of f over D by

$$\iint_D f(x,y)\,dA = \iint_R g(x,y)\,dA = \iint_R g(x,y)\,dx\,dy$$

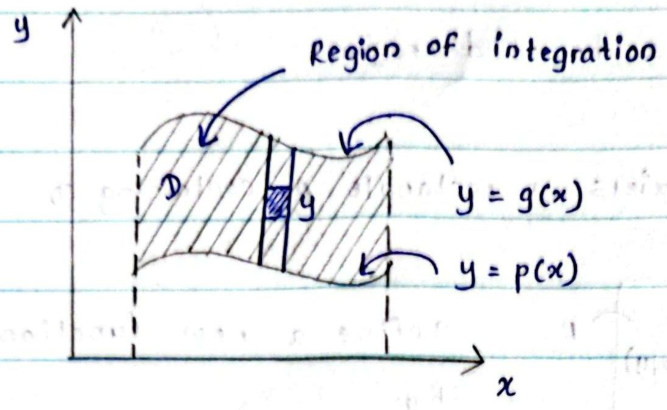


$R = D \cup D'$

$$\iint_R g(x,y)\,dA = \iint_D g(x,y)\,dA + \iint_D g(x,y)\,dA$$

$$= {}^{D}\iint_D f(x,y)\,dA\ D'$$

Typ 1



$D = \{(x, y) \in \mathbb{R}^2 \mid p(x) \le y \le q(x) \ , \ a \le x \le b\}$

we may prove that

$$\iint_D f(x, y)\, dA = \int_{x=a}^{b} \left[\int_{y=p(x)}^{q(x)} f(x, y)\, dy \right] dx$$

Example 3

let D be the region bounded by the parabolas: $y = 2x^2$ and $y = 1 + x^2$

Evaluate $\iint_D (3x + 6y)\, dA$ . . .

let $y_1 = 2x^2$ and $y_2 = 1 + x^2$

$$y_1 = y_2 \iff 2x^2 = x^2 + 1$$
$$x^2 = 1$$
$$x = \pm 1$$

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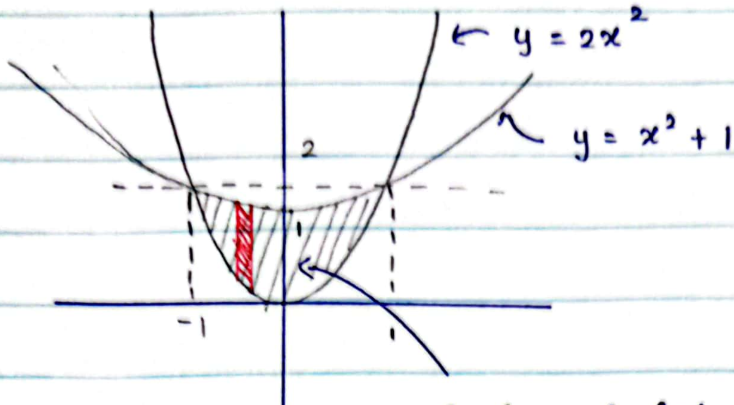
That is, the intersection points are : $(1,2)$ and $(-1,2)$



Region of integration : Type I

Now,

$$\iint_D (3x+6y)\,dA = \int_{x=-1}^{1} \left[\int_{y=\cancel{2}x^2+1}^{2x^2 \cancel{+1}} (3x+6y)\,dy \right] dx.$$

$$= \int_{x=-1}^{1} \left[3xy + 3y^2 \right]_{2x^2}^{x^2+1} dx$$

$$= \int_{x=-1}^{1} -3x(x^2+1) + 3x(2x^2) - 3(x^2+1)^2 + 3(2x^2)^2$$

$$= \int_{x=-1}^{1} -3x^3 - 3x + 6x^3 - 3x^4 - 6x^2 + 3 + 12x^4\,dx$$

$$= \int_{x=-1}^{1} +9x^4 + 3x^3 - 6x^2 - 3x - 3\,dx$$

$$= \cancel{-9/5\,x^5} -$$

$$= 3\int_{x=-1}^{1} 3x^4 + x^3 - 2x^2 - x - \cancel{1}\,dx.$$

$$= 3\left[\tfrac{3}{5}x^5 + \tfrac{1}{4}x^4 - \tfrac{2}{3}x^3 - \tfrac{1}{2}x^2 - \cancel{1}x \right]_{-1}^{1}$$

$$= 3\left[\tfrac{3}{5} + \tfrac{3}{5} + \tfrac{1}{4} - \tfrac{1}{4} - \left(\tfrac{2}{3} + \tfrac{2}{3}\right) - \left(\tfrac{1}{2} + \tfrac{1}{2}\right) - (1+1) \right]$$

$$= 3\left[\tfrac{6}{5} - \tfrac{4}{3} - 2 \right]$$

$$= \cancel{3} \times \frac{18-20-30}{\cancel{15}\,5} = \frac{-32}{5}$$